Introduction to stability and control problems

ELEC0447 - Analysis of electric power and energy systems

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Overview

1. The concept of stability

2. Why do we need stability?

Why control power systems?

- Technical requirements: power system devices are designed so as to operate within well-defined "tolerance regions"
  - around nominal values of voltage V_n: 1 ± 0.1 pu in Europe
  - around nominal value of frequency f_n: 50 ± 0.2 Hz in Europe (in steady state)
  - within the P-Q capabilities of devices
  - under the current limits of lines and transformers
- Large/persistent deviations from nominal values could lead to
  - damages and safety problems (e.g. high voltage)
  - cascading phenomena
  - service interruptions

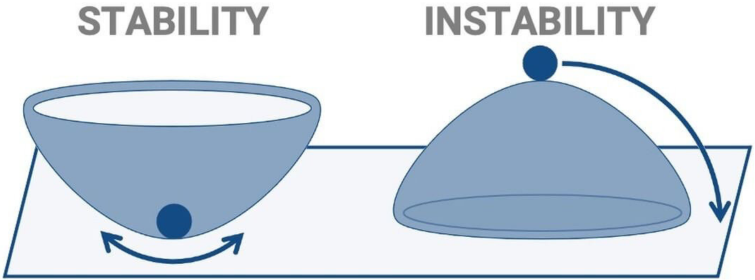
Exogenous threats

- Sudden disturbances, such as line or generator tripping
- Fast variations of the *net load* (cf. Duck curve)
  - the net load refers to the load "seen" by the transmission system, i.e. the load minus the non-controllable dispersed generation
- weather conditions, such as storms, which can impact the generation of renewable energy sources (RES) (e.g. wind turbines' cut-out speed)

The concept of stability

General concept

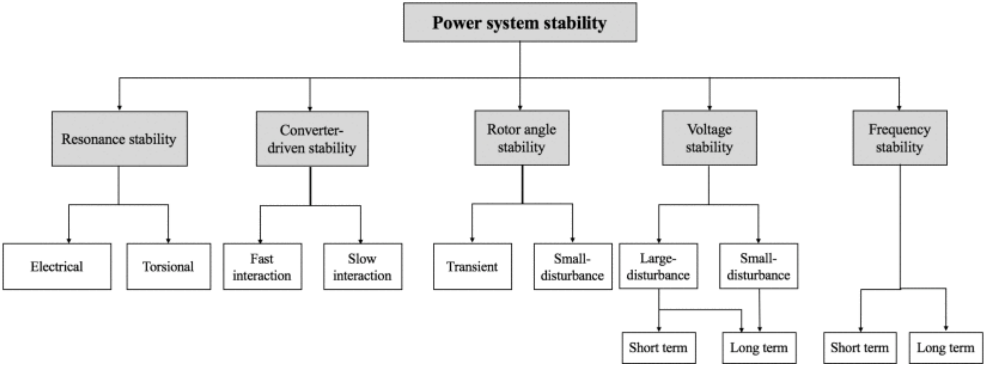
If a system has the property that it will get back into the equilibrium state again after moving away from its equilibrium state, then it is stable. [1]



In power systems I

Power system stability is the ability of an electric power system, for a given initial operating condition, to regain a state of operating equilibrium after being subjected to a physical disturbance, with most system variables bounded so that practically the entire system remains intact. *[2]*

In power systems II



Why do we need stability?

Key points

- We often take electricity as a simple commodity.
- But the electric power system is one of the most complex and largest man-made systems.
- The chances of system failures are very high taking into account the impact of external factors and rapid changes in the system's state.
- However, power systems are very reliable (operated 24h/24h 7d/7d and only a few hours of power outages per year!).
- But when instabilities occur, it can lead to blackouts with huge financial and societal consequences.

Tokyo 1987

- Unexpected load increase and presence of constant power devices (air conditioners) led to a voltage collapse.
- **8 GW lost** and **2.8M customers impacted** [3, 4]

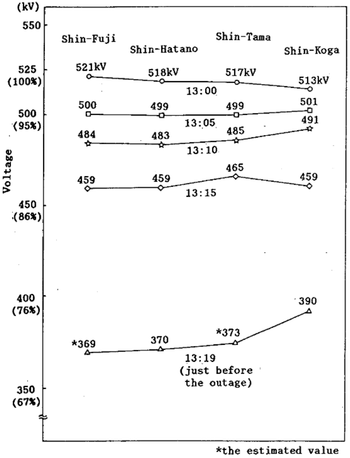
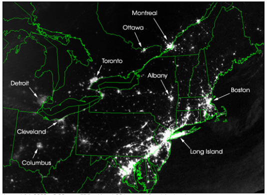


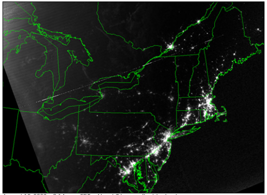
Fig.3 Voltage Drops at Main Substations

Canada/Northeast USA 2003

- Initial problem: not enough reactive power reserve. Hot weather and large consumption led to transmission lines overloading and eventually sagging into trees, further deteriorating the initial problem.
- A cascading event caused the tripping of hundreds of lines and generating units.
- **63 GW lost** and **50M customers impacted** and **Estimated cost above $5 Billion**
- Watch `https://www.youtube.com/watch?v=nd3teNgUq8E`, `https://www.youtube.com/watch?v=KciAzYfXNwU`



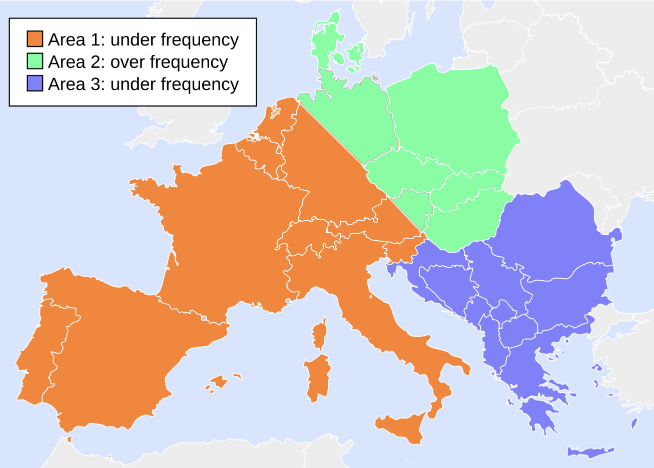
August 14, 2003 • 9:29 p.m. EDT • About 20 hours before blackout


August 15, 2003 • 9:14 p.m. EDT • About 7 hours after blackout

Europe 2006 I

- Disconnection of a transmission line in Germany for the transport of a ship approved by the local TSO.
- The local TSO approved to advance the disconnection later that day, but the commercial flows remained unchanged.
- Some lines were critically loaded because of the line disconnection and a fast increase of load consumption led to a cascading event.
- The European interconnected network was split into 3 islands.
- Watch `https://www.youtube.com/watch?v=A30DdnsICuw`
- **14.5 GW lost** and **15M customers impacted** [5]

Europe 2006 II



Brazil 2023

- False operation of a relay protection system led to a 500kV line disconnection.
- The Energy Management System did not operate properly.
- **19 GW lost**

Conclusion

- Every time a blackout happened, lessons have been learned, and new rules have been put in place.
- Nonetheless, due to the system complexity, new complex phenomena occur that power system engineers try to understand.
- In the following, we'll dive into the mechanisms of voltage instability.

References I

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